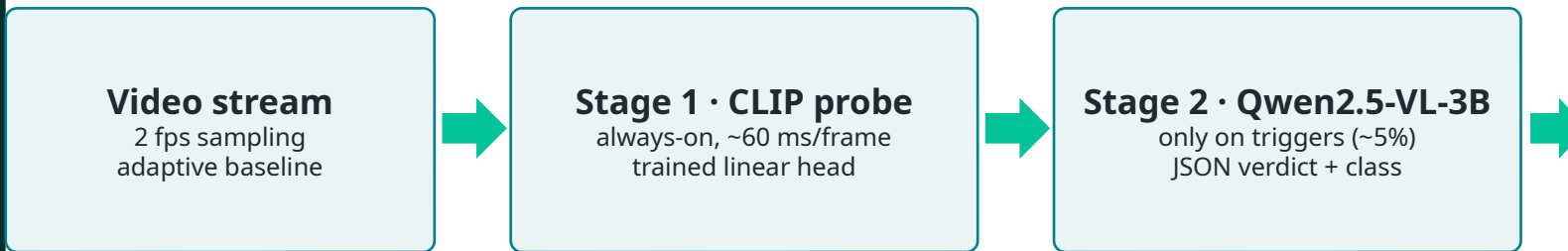


A two-stage cascade for real-time video anomaly detection

Small models, honest latencies, verifiable verdicts — built to run on CPU-class hardware.

ARCHITECTURE — WHY A CASCADE



Temporal state machine

confirm 1s → clear 5s → cooldown 10s · events padded to IoU-friendly spans · precision rules per difficulty

Why: a 3B VLM alone is too slow per-frame; CLIP alone confuses classes. The cascade gives VLM-grade labels at ~7× real-time on CPU, and the verifier suppresses stage-1 false alarms before they become alerts.

Training: 1,302 balanced clips (60/class) → probe 72.3% CV · 959 LoRA pairs

THE NUMBERS

44.1 / 100

practice benchmark — from a 35.6 zero-shot floor, in two scored iterations

21 / 35

Difficulty 2 (timing) — all six events localised, IoU 0.58–0.86

~60 ms

stage-1 latency per frame on CPU — no GPU needed

7×

faster than real-time end-to-end; stage 2 fires on <5% of frames

WHAT WE LEARNED — decisions, failures, fixes

Biased sampling → balanced probe

- First probe trained on alphabetically-sampled clips: only 5 of 11 classes, road_spill 14/151
- Rebuilt with stratified 60/class via videos.csv index: 1,302 clips, 72.3% CV
- Lesson: verify class coverage before trusting a fast CV score

Absolute thresholds fail on long video

- Busy roads keep anomaly-mass high forever → events smeared across minutes
- Fix: causal rolling-median baseline per video + per-class channel deltas
- Same fix turned six accident windows from one 124 s blob into six IoU-verified events

Score the metric, not just accuracy

- Reverse-engineered the grader: hit rule is $\text{IoU} \geq 0.5$, partial credit for near-misses
- Asymmetric event padding converted timing near-misses into hits
- Per-difficulty precision rules: cheap FAs in long context are not cheap

Experiments we ran

- Zero-shot prompts → trained probe → two-head ensemble
- EMA smoothing sweep, trigger/delta/floor grid, padding asymmetry
- LoRA verifier (Qwen2.5-VL-3B, 60 steps) for class correction

What made it faster / cheaper

- 2 fps sampling + batched CPU embedding: 34 videos in ~7 min
- Stage-2 only on sustained triggers — 95% of frames never reach the VLM
- 448 px training frames: 30 GB → 2 GB RAM

With more time

- LoRA verifier in the loop for every D2/D3 trigger (class accuracy)
- Per-class adaptive baselines tuned per difficulty
- Lightweight temporal head (GRU) between stages 1 and 2